Constraints

Compaction models geometry as a directed constraint graph

The idea

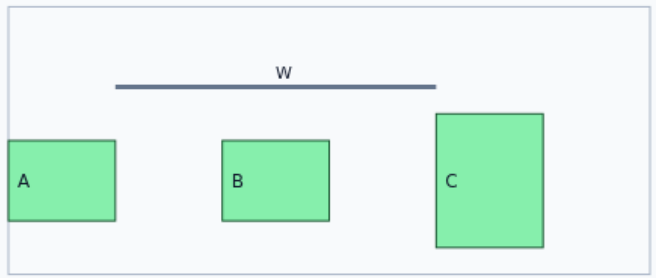
- Nodes L_A,R_A,…; size edges R≥L+width; spacing edges L_right ≥ R_left + gap
- GOLDEN ~10 X edges

Nodes

CONSTRAINT GRAPH

Nodes

Step 1 / 5 · nodes · module02-01-constraint-graph



Explanation

Left/right edge nodes per object.

- L_/R_

```
vars
```

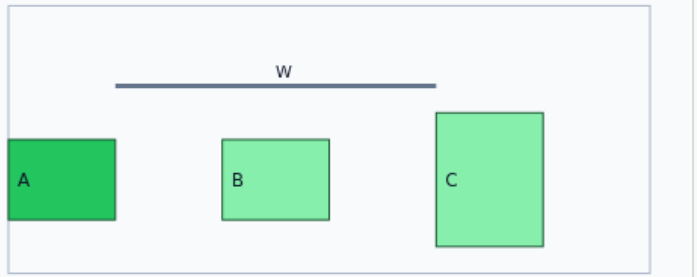
Nodes

Size

CONSTRAINT GRAPH

Size

Step 2 / 5 · size · module02-01-constraint-graph



Explanation

$R_{id} \geq L_{id} + \text{width}(id)$.

- size edge

```
width
```

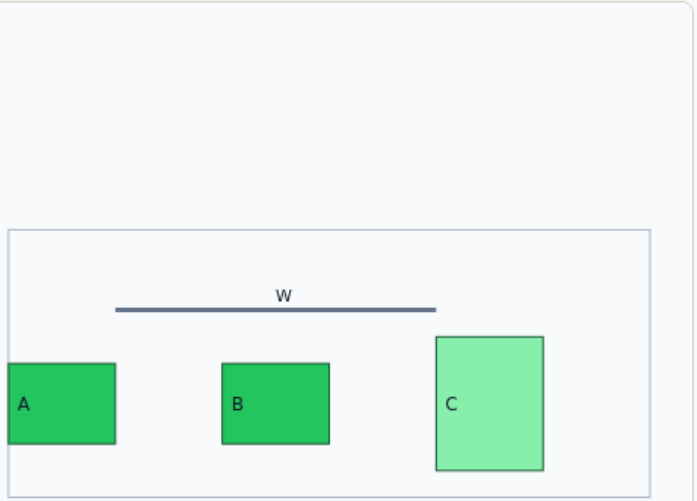
Size

Spacing

CONSTRAINT GRAPH

Spacing

Step 3 / 5 · space · module02-01-constraint-graph



Explanation

Next cell's left ≥ previous right + gap.

- gap edge

```
min_gap
```

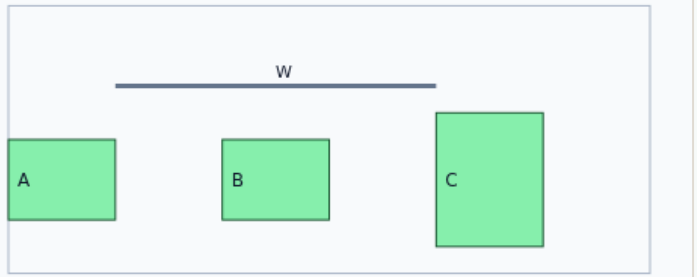
Spacing

Source

CONSTRAINT GRAPH

Source

Step 4 / 5 · src · module02-01-constraint-graph



Explanation

SRC_X anchors the left of the chip.

- anchor

```
0
```

Source

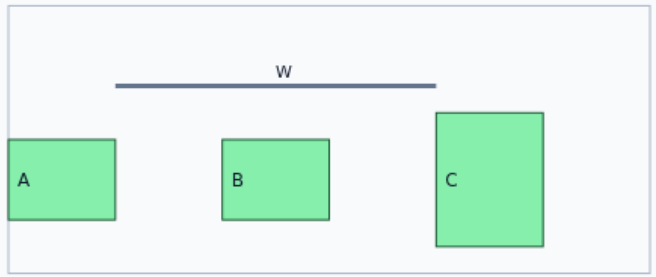
Count

CONSTRAINT GRAPH

Count

Step 5 / 5 · count · module02-01-constraint-graph

edges=10



Explanation

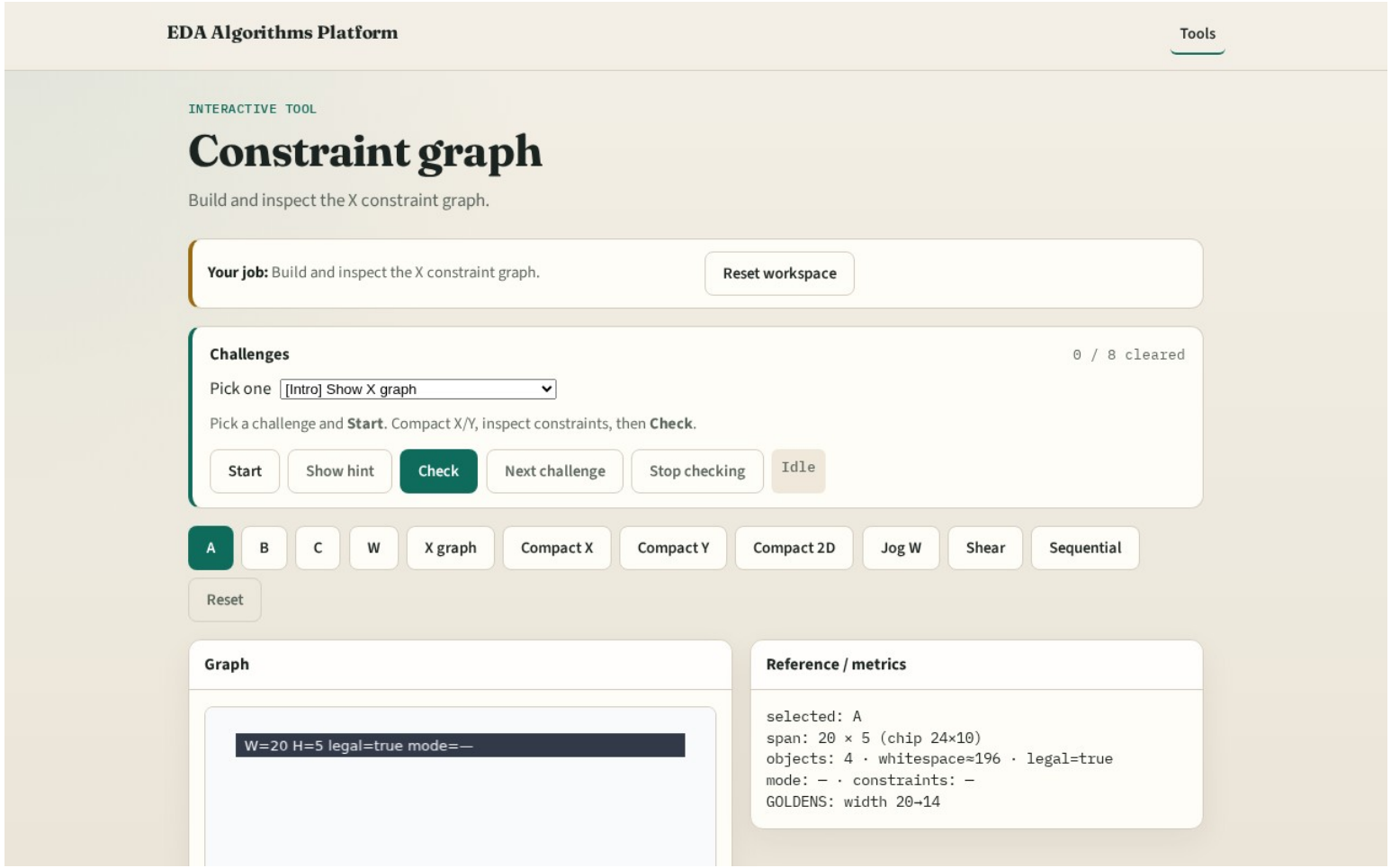
GOLDEN: about 10 constraint edges on X.

- |E|

≥8

Count

Browser lab track



Browser lab starter

Implement track

- Call build_constraint_graph(DATA,'x'); print len(edges)

Pitfalls

- Forgetting size edges
- Mixing X and Y nodes

Your turn

- Next: longest-path placement